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B. E. (Third Semester) Examination, Nov.-Dec., 2015

(New Scheme)

(Elect. Engg. Branch)

ELECTRICAL MACHINES–I

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) is compulsory and carries 2 marks. Attempt any two parts from (b), (c) and (d) each carry 7 marks.*

Unit-I

1. (a) Explain how core loss can be minimized in single phase transformer?

(b) Draw the phasor diagram of single phase transformer for lagging p.f. load with the help of equivalent circuit diagram.

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(c) A 1 phase 250/500 V transformer gave the following results,

Open circuit test 250 V, 1 Amp, 80 Watt on LV side

Short cirtuit test 20 V, 12 Amp, 100 Watt on HV side

Calculate the circuit constants and show them on equivalent circuit.

(d) What is voltage regulation of transformer? For which type of load voltage regulation is negative, derive the expression using equivalent circuit diagram?

Unit-II

2. (a) Write down at least two differences between Power transformer and distribution transformer?

(b) Derive an expression for saving of copper in auto-transformer as compared to an equivalent 2 winding transformer. Also draw and explain phasor diagram of Step up auto transformer.

(c) Describe the Back to Back test of determining the regulation and efficiency of a pair of similar transformers, giving the circuit diagrams and indicating what reading will be necessary.

(d) The daily variation of load on a 100 kVA transformer is as follows :

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8:00 am to 1:00 pm	65 kW	35 kVAr
1:00 pm to 6:00 pm	80 kW	50 kVAr
6:00 pm to 1:00 am	30 kW	30 kVAr
1:00 am to 8:00 am	no load	

This transformer has no load core loss of 270 watts and full load ohmic loss of 1,200 watts. Detrmine the all day efficiency of transformer.

Unit-III

3. (a) What do you understand by bank in case of 3 phase transformer?

(b) Explain why parallel operation of transformer is required and State necessary conditions for parallel operation of 3 phase transformer?

(c) Draw and explain the circuit diagram of a transformer arrangement for converting from three phase to two phase supply.

(d) A 3 phase step down transformer is connceted to 6,600 V mains and takes 10 Amp. The ratio of turns per phase is 12. Neglect losses. Calculate the secondary line current, line voltage and output power for following connections.

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(i) Y-y

(ii) Y-d

(iii) D-y

Unit-IV

4. (a) Explain the function and meterial used for :

(i) Yoke

(ii) Interpole core

(b) What is Bill concept and derive an equation for electro-magnetic torque in DC machine.

(c) A 4 pole generator has wave wound armature with 722 conductors, and its deliver 100 A on full load. If the brush lead is 8 degree mechanical form the G. N. A. Calculate the artmature demagnetizing and cross magnetizing ampere turns per pole?

(d) A 6 pole shunt generator runs at 1000 rpm and has following data armature resistance is $1\,\Omega$ field resistance is $120\,\Omega$ and number of conductors is 350, wave wound. Flux per pole is 0.02 weber, load resistance is $30\,\Omega$. Find load current.

Unit-V

5. (a) Why dc motors draw heavy current at starting?

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(b) Explain various types of losses occurring in DC machines.

(c) A 4 pole, 250 V wave connected shunt motor gives 10 kW when running at 1000 rpm and drawing armature and field currents of 60 A and 1 A respectively. It has 560 conductors. Its armature resistance is 0·02 Ω assuming a drop of 1 V per brush, determine (a) total torque (b) useful torque (c) useful flux per pole (d) rotational losses (e) efficiency.

(d) Explain the Hopkinson's test of D.C. machine.

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